

Actomyosin remodeling regulates biomineral formation, growth and morphology during eukaryote skeletogenesis

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1
September 7, 2023 (this version)

Posted to preprint server
May 26, 2023

Sent for peer review
May 25, 2023

Eman Hijaze, Tsvia Gildor, Ronald Seidel, Majed Layous, Mark Winter, Luca Bertinetti, Yael Politi, Smadar Ben-Tabou de-Leon 

Department of Marine Biology, Leon H. Charney School of Marine Sciences, University of Haifa, Haifa 31905, Israel
• B CUBE Center for Molecular Bioengineering, Technische Universität Dresden, 01309 Dresden, Germany •
Department of Electrical Engineering, Computer Science and Mathematics, Technische Universiteit Delft, Delft 2628CD, Netherlands

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Biomineralization had apparently evolved independently in different phyla, using distinct minerals, organic scaffolds and gene regulatory networks (GRNs). However, diverse eukaryotes from unicellular organisms, through echinoderms to vertebrates, use the actomyosin network during biomineralization. Specifically, the actomyosin remodeling protein, Rho-associated coiled-coil kinase (ROCK) regulates cell differentiation and gene expression in vertebrates' biomineralizing cells, yet, little is known on ROCK's role in invertebrates' biomineralization. Here we reveal that ROCK controls the formation, growth and morphology of the calcite spicules in the sea urchin larva. ROCK expression is elevated in the sea urchin skeletogenic cells downstream of the Vascular Endothelial Growth Factor (VEGF) signaling. ROCK inhibition impairs the organization of F-actin around the spicules, disrupts skeletogenic gene expression and leads to skeletal loss. ROCK inhibition after spicule formation reduces spicule elongation rate and induces ectopic spicule branching. Reduced skeletal growth and enhanced branching are also observed under direct perturbations of the actomyosin network. Similar skeletogenic phenotypes are observed when ROCK is inhibited in a skeletogenic cell culture, indicating that these phenotypes are due to ROCK activity specifically in the skeletogenic cells. We propose that the actomyosin network was employed independently, downstream of distinct GRNs, to regulate biomineral growth and morphology across Eukaryotes.

eLife assessment

This **valuable** paper addresses an **important** problem, the cell biological control of biomineralization, with the sea urchin embryo as an experimental model. The paper provides evidence that ROCK and the cytoskeleton play a role in biomineralization, but the evidence is deemed currently **incomplete**, as there are concerns regarding the efficacy and specificity of the reagents used to perturb ROCK function. In addition, and more importantly, the data do not point to a plausible mechanism by which the actin cytoskeleton might regulate the biomineralization process.

Introduction

Biomineralization is the process in which organisms from the five kingdoms of life, use minerals to produce shells, skeletons and teeth that protect and support them (1–4). Recent studies suggest that biomineralization evolved by the phylum specific co-option of ancestral gene regulatory networks (GRNs) that drove the construction of distinct organic scaffolds (4–9) and by the evolution of specialized sets of biomineralization proteins (3, 7, 10–13). This explanation is in line with the dissimilar GRNs and biomineralization proteins that drive biomineralization in different phyla (5, 12, 14). Yet, there are common cellular processes required for mineralization in highly diverse phyla (4), suggesting that distinct upstream GRNs might have recruited common cellular and molecular mechanisms to drive biomineralization.

The actomyosin network was shown to play a role in biomineralization in various Eukaryote models, from unicellular organisms to vertebrates' bones and teeth (15–22). There is a tight association between actin filaments and the biomineralization compartment in diatoms (19) and a highly dynamic actin organization that forms the biomineralization compartment in foraminifera (17). Perturbations of actin polymerization result with severely deformed shells and inhibition of shell shedding in coccolithophores (15, 16) and diatoms (18). In vertebrates, Rho GTPases and Rho56 associated coiled-coil kinase (ROCK), regulate chondrocytes, osteoblasts and odontoblasts differentiation and affect genes expression in these biomineralizing cells (20–22). However, the roles of actomyosin remodeling in controlling mineral deposition and shaping biomineral morphology, are still unclear.

Sea urchin larval skeletogenesis provides an attractive model to study the role of the actomyosin network in biomineralization. Sea urchin larval skeletons are made of two frameworks of interconnected calcite rods, termed “spicules” that are generated by the skeletogenic cells (6, 23). To make the spicules, the skeletogenic cells form a ring with two lateral skeletogenic clusters and fuse through their filopodia forming a pseudopodia cable that links them into a syncytium (24, 25). The mineral is concentrated in the form of amorphous calcium carbonate (ACC) inside intracellular vesicles (26, 27). The vesicles are secreted into the biomineralization compartment generated in the skeletogenic cell clusters forming two triradiate spicules (Fig. 7A, (26, 28)). The tubular biomineralization compartment, also called the spicule cavity, elongates within the pseudopodia cable by localized mineral deposition at the tip of each rod (29).

The GRN that controls sea urchin skeletogenesis is known in great detail and is highly similar to the GRN that controls vertebrates' vascularization, suggesting a common evolutionary origin of these two tubulogenesis programs (6, 23, 30). As the spicules elongate, the expression of key regulatory and biomineralization related genes becomes restricted to the skeletogenic cells proximal to the growing tips, possibly to promote mineral deposition at these sites (31–33).

Localized gene expression is regulated by signaling cues such as the Vascular Endothelial Growth Factor (VEGF) signaling (31 [↗](#), 32 [↗](#)). However, how the skeletogenic GRN drives spicule formation and localized mineral deposition at the tips of the rods is poorly understood.

Previous works suggest that the actomyosin network is essential for sea urchin skeletal growth. Calcium bearing vesicles perform an active diffusion motion in the skeletogenic cells with a diffusion length that inversely correlates with the strength and activity of the actomyosin network (34 [↗](#)). Actin filaments are formed around the spicule (34 [↗](#), 35 [↗](#)) and F84 actin signal is enriched at the tips of the growing skeletal rods in skeletogenic cell culture (36 [↗](#)). Genetic and pharmacological perturbations of the GTPase, CDC42, prevent the formation of filopodia in the skeletogenic cells and inhibit spicule formation and elongation (37 [↗](#)). Pharmacological perturbations of ROCK prevent spicule formation (38 [↗](#)) and genetic perturbations of Rhogap24l/2 result with ectopic spicule splitting (6 [↗](#)). Despite these phenotypic characterizations, little is known on the regulation and role of the actomyosin network and remodeling proteins in localized mineral deposition.

Here we study the role of ROCK and the actomyosin network in the sea urchin *Paracentraorus lividus* (*P. lividus*). Our findings reveal the critical role of ROCK and the actomyosin network in multiple aspects of sea urchin biomineralization, suggesting a common use of these factors in Eukaryote biomineralization, downstream of distinct GRNs.

Results

ROCK is enriched in the skeletogenic cells depending on VEGF signaling

We sought to study the spatial expression of the ROCK protein and its regulation by VEGF signaling, a prominent regulator of sea urchin skeletogenesis (6 [↗](#), 30 [↗](#)). The sequence of ROCK and especially, its functional domains are highly conserved between human and sea urchin (Fig. S1, (39 [↗](#), 40 [↗](#))). We therefore used a commercial antibody that recognizes human ROCK protein to test ROCK expression in *P. lividus* embryos (Fig. S2). We first used western blot to detect ROCK expression in control and under VEGFR inhibition at the time of spicule formation and during skeletal elongation (~22 hours post fertilization [hpf], 27hpf and 33hpf, Fig. S2). The antibody detected a ~150kDa protein (Fig. S2A) that is the predicted size for *P. lividus* ROCK protein (153kDa). VEGFR inhibition marginally increased the overall level of ROCK at 22hpf, but did not affect it at 27hpf and 33hpf (Fig. S2B). Yet, this measurement was done on proteins extracted from whole embryos, and we wanted to study the spatial expression of ROCK and specifically, its regulation in the skeletogenic cells.

We studied the spatial distribution of ROCK protein using ROCK antibody and the skeletal cell marker, 6a9, that marks the skeletogenic syncytium membrane (41 [↗](#)), **Fig. 1** [↗](#)). We quantified ROCK signal in the skeletogenic cells compared to the neighboring ectodermal cells in control embryos and under VEGFR inhibition, at the three time points mentioned above. At 22hpf, ROCK expression in the skeletogenic cells is mildly enriched compared to the neighboring ectodermal cells and this enrichment is not observed under VEGFR inhibition (**Fig. 1A-F** [↗](#)). At 27hpf, ROCK enrichment in the skeletogenic cells increases, but similar enrichment is observed under VEGFR inhibition (**Fig. 1G-L** [↗](#)). At 33hpf, ROCK enrichment in the skeletogenic cells is most apparent (**Fig. 1M-O** [↗](#)) and depends on VEGF signaling (**Fig. 1P-S** [↗](#)). At both 27hpf and 33hpf in control embryos, ROCK seems localized near the skeletogenic cell membranes, which further supports its activation in these cells, since ROCK activation leads to its localization to the cell membranes in other systems (42 [↗](#)). Overall, this data demonstrate that ROCK expression is elevated in the skeletogenic cells and this enrichment strengthens with skeletal elongation and depends on VEGF signaling.

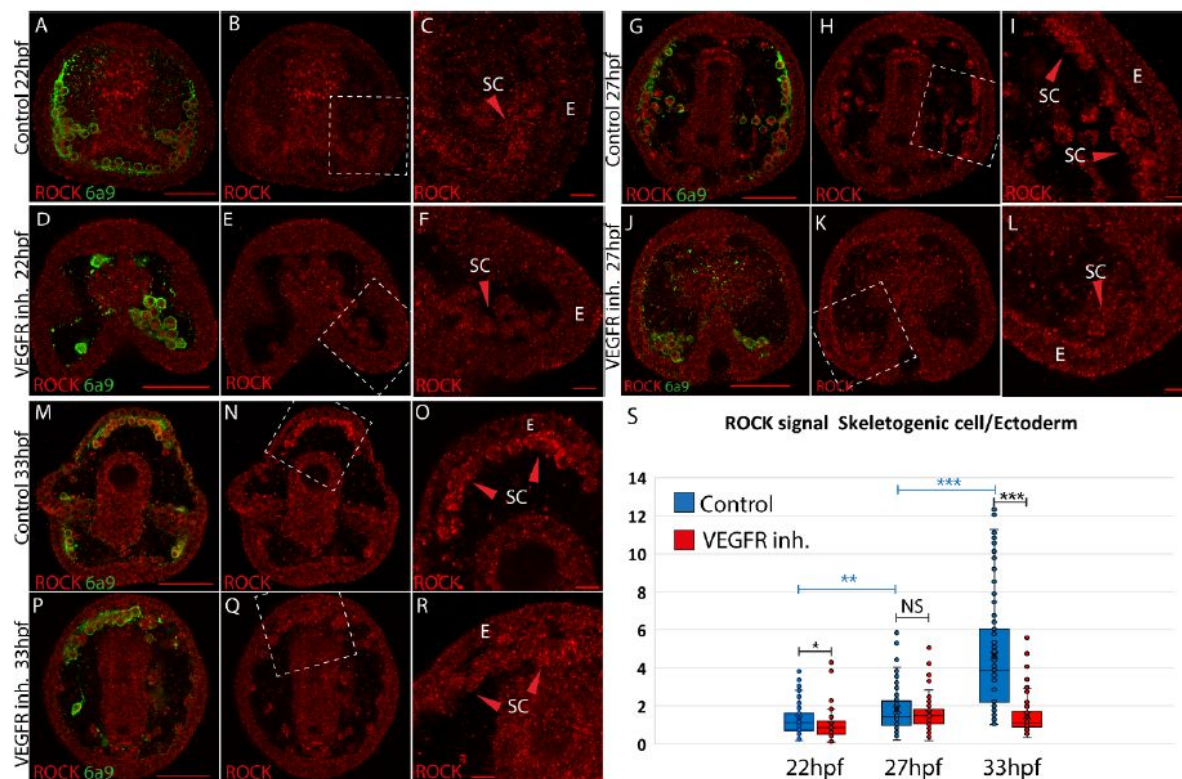


Figure 1

ROCK enrichment in the skeletogenic cells increases with time and depends on VEGF signaling

A-R, ROCK immunostaining at different times points in control and under VEGFR inhibition. In each condition and time point, the left image shows ROCK immunostaining and the skeletogenic cell marker, 6a9 (A, D, G, J, M and P). The middle image shows ROCK immunostaining alone in the whole embryo (B, E, H, K, N and Q). The right image shows the enlargement of the white rectangle region of the middle image (C, F, I, L, O, R). Scale bar in whole embryo is 50µm and in enlargements is 10µm. E – ectoderm, SC – skeletogenic cells. S, quantification of the ratio between ROCK signal/area in the skeletogenic cells compared to the ectodermal cells (see methods for details). Each box plot shows the average marked in x, the median, the first and the third quartiles (edges of boxes) and all experimental measurement (dots). Experiments were performed in three independent biological replicates and in each condition at least 33 embryos were measured. Statistical significance was measured using paired 2-tailed t-test where, * indicates $P<0.05$, ** indicates $P<0.005$, and *** indicates $P<0.0005$.

ROCK activity in the skeletogenic cells controls spicule initiation, growth and morphology

Next, we studied the role of ROCK in sea 128 urchin skeletogenesis using genetic perturbations. We downregulated the expression of ROCK by the injection of two different translation morpholino anti-sense oligonucleotides (MASOs). ROCK MASO-1 matches the start of translation region, ROCK MASO-2 matches the 5' region upstream the start of translation, and random MASO was used as a control. Embryos injected with either ROCK MASO-1 or MASO-2 show strongly reduced skeletons (**Fig. 2C, D**) and embryos injected with ROCK MASO-2 show an additional phenotype of severely branched skeleton (**Fig. 2B**). The additional branching phenotype and the larger percentage of affected embryos in ROCK MASO-2 are probably due to its higher efficiency resulting from its more 5' location relative to translation start codon (**Fig. 2E**). Thus, the genetic perturbations of ROCK expression indicate that sea urchin ROCK is important for skeletal elongation and normal branching.

To elucidate the dependence of ROCK phenotypes on ROCK activity level and identify its effect on different stages of skeletogenesis, we tested the skeletogenic phenotypes of different concentrations of ROCK inhibitor, Y27632, applied before or after spicule initiation (**Fig. S3A**). Y27632 binds to the ATP site of ROCK's kinase domain and prevents its activity (43). The amino-acids to which Y27632 binds, are conserved in the sea urchin ROCK protein, supporting its specific inhibition of this protein (**Fig. S1C**). Continuous ROCK inhibition, beginning at egg fertilization, using 80µM Y27632, did not affect skeletogenic cell migration but eliminated skeletal formation at 27hpf (**Fig. S3B, C**). At two days post fertilization (dpf) the majority of the embryos completely lack skeleton and the rest show only spicule initiations, yet their overall embryonic development seems normal (**Fig. 2G**, **S3I**, in agreement with (38)). Similar results are observed when adding the inhibitor in this concentration at 20hpf (**Fig. 2I**, **S3I**). Yet, the addition of 80µM of the inhibitor at 25hpf, after spicule initiation, results in a majority of embryos with severe ectopic spicule branching, which resembles the branching phenotype observed in ROCK MASO-2 (**Fig. 2H**, **I**). Reducing the concentration of the added inhibitor leads to milder branching and elongation phenotypes (**Fig. S3F, H, K**). However, even at the lowest inhibitor concentration, we detect a reduction of skeletal growth rate and ectopic branching a few hours after the addition of ROCK inhibitor (**Fig. S4**, 30µM). Washing the inhibitor after 25hpf results in partial recovery of skeletogenesis, with normal or mildly delayed skeletons (**Fig. S3E, K**). Overall, our genetic and pharmacological perturbations indicate that ROCK activity is necessary for spicule formation, skeletal elongation and normal branching pattern.

To test if ROCK skeletogenic phenotypes are due to its activity specifically in the skeletogenic cells, we inhibited ROCK in a culture of isolated skeletogenic cells (**Fig. 2J-N**). The addition of ROCK inhibitor to the skeletogenic cell culture at 16hpf, before the spicules form, completely abolished spicule formation (**Fig. 2K**). The addition of the inhibitor at 48hpf, after spicule formation, resulted with significantly shorter spicules (**Fig. 2L, N**) and increased branching compared to control spicules (**Fig. 2M**). The reduction of the spicule length becomes more significant with increasing concentration of the ROCK inhibitor (**Fig. 2N**). Overall, ROCK skeletogenic phenotypes in the isolated skeletogenic cell culture are similar to its phenotypes in whole embryos, verifying that its specific activity in the skeletogenic cells is essential for spicule formation, normal elongation and prevention of branching.

ROCK inhibition reduces spicule volume, surface area and total length, but not thickness (SR-µCT)

To quantify the effect of ROCK inhibition on mineral deposition and skeletal shape we used synchrotron radiation micro-computed tomography (SR-µCT) that enables three-dimensional visualization and quantification of spicule morphology at 2 and 3dpf (**Figs. 3**, **S5**, see methods for details). The aberrant morphology of ROCK inhibited spicules is visible at both developmental

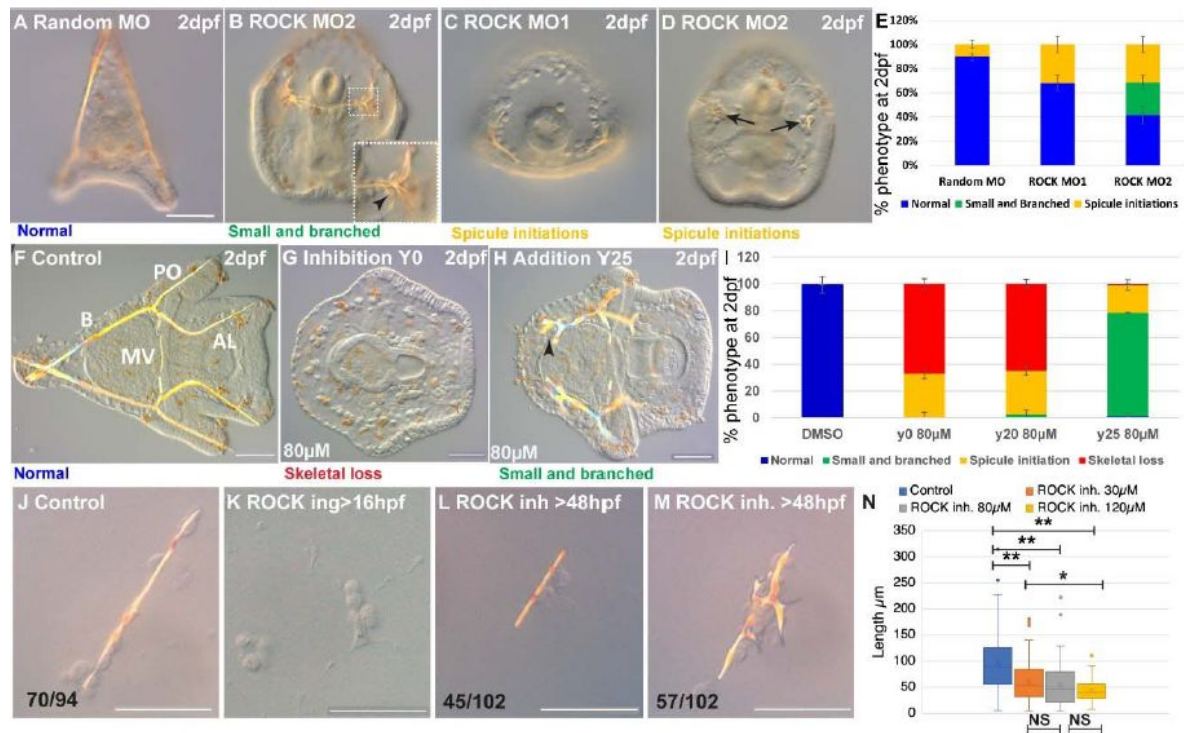


Figure 2

ROCK activity is essential for spicule formation, normal elongation and branching in whole embryos and in skeletogenic cultures

A-E, genetic perturbation of ROCK translation using two different MASOs observed at 2dpf. A, control embryo injected with Random MASO. B, embryo injected with ROCK MO-2 shows ectopic spicule branching. C, D, embryos injected with ROCK MO-1 or MO-2 show spicule initiations. E, summary of MASO injection phenotypes based on 4-6 independent biological replicates. F-I, pharmacological perturbations of ROCK activity using 80μM of the inhibitor Y27632 observed at 2dpf. F, representative control embryo with normal skeletal rods, B, body; AL, anterolateral; PO, post-oral and MV, midventral. G, complete skeletal loss in embryo treated continuously with ROCK inhibitor. H, reduced skeletal growth and enhanced ectopic branching in embryo where ROCK inhibitor was added at 25hpf. I, summary of perturbation phenotypes based on 3-8 biological replicates for each treatment. See additional phenotypes and summary in Fig. S3 and Table S1. J-M, representative spicules from skeletogenic cell cultures in control and under 30μM Y27632 at 72hpf. J, linear spicule in control culture, K, Y27632 addition at 16hpf, before spicule initiation, completely blocks spiculogenesis. L, Y27632 addition after spicule initiation, at 48hpf, reduces spicule elongation, M, and enhances branching. N, Quantification of spicule length in control and ROCK inhibition (>48hpf) at 72hpf. *P < 0.05, **P < 0.001, Kruskal-Wallis non-parametric test. Results are based on three biological repeats for each treatment, except from 120μM that was done in two biological repeats. Scale bars are 50μm. In J, L, M, the numbers at the bottom indicate the number of spicules that show this phenotypes (left) over all observed spicules (right).

time points (**Fig. 3A**). The ectopic branching occurs both at the tips where tip-splitting is detected (arrowheads, **Fig. 3A**), as well as in the back, where small spicule extensions, or mineral ‘dripping’ are observed (arrow at 48hpf, **Fig. 3A**). This implies that the regulation of mineral deposition is perturbed both at the tip and in the back of the spicules under ROCK inhibition.

When compared to control spicules, ROCK inhibition leads to a ~2.5-fold reduction of the spicule volume and surface area in both 2 and 3dpf (**Fig. 3B-D**, see Table S2 for measured values and standard deviations and S3 for statistical analyses). These SR-μCT measurements show that the rate of mineral deposition is significantly reduced under ROCK inhibition.

To understand the specific effect of ROCK inhibition on spicule elongation vs. thickening, we compared the total length and the mean thickness of the spicules between control and ROCK inhibition (**Fig. 3E-G**). The total length of control spicules was on average about 2.5-times longer than the length under ROCK inhibition in both 2dpf and 3dpf (**Fig. 4E, F**, tables S2 and S3). In contrast to its effect on the spicule length, ROCK inhibition caused a minor increase of the spicule thickness at 2dpf and at 3dpf the thickness was not affected by the inhibition (**Fig. 3E, G**, Tables S2, S3). These SR-μCT measurements indicate that ROCK activity plays a crucial role in controlling the rate of mineral deposition at the tips of the rods and in spicule elongation, and is not affecting spicule thickening.

ROCK activity does not control mineral uptake but is required for “tip-dominance”

To test the effect of ROCK inhibition on mineral intake, we used the fluorescent chromophore, calcein that binds to divalent ions including Ca^{+2} and is widely used to track the calcium pathway in biomineralization (6, 28, 33, 34). We measured the number of calcein stained vesicles per area as well as the number of pixels of calcein signal per area in the skeletogenic cells, in control and under ROCK inhibition at 1dpf and 2dpf (**Fig. S6A-E**). Continuous ROCK inhibition does not change the number of calcium bearing vesicles at 1dpf, but the number of calcein stained pixels significantly increases in this condition, possibly indicating that the vesicle volume is larger (**Fig. S6F, G**). At 2dpf, the number of calcein stained pixels is similar in control and in continuous ROCK inhibition, suggesting that the overall calcium content does not change in this condition (**Fig. S6H, I**). The number of calcein stained vesicles is however decreased under ROCK inhibition, possibly indicating that there are fewer vesicles with larger volume. Addition of ROCK inhibitor at 25hpf affects neither the number of calcein stained vesicles nor the number of calcein stained pixels in the skeletogenic cells. Together, these measurements show that ROCK is not required for the uptake of calcium into the cells. Therefore, the skeletal phenotypes occurring under ROCK inhibition are not related to a decrease in calcium uptake or shortage in intracellular calcium, but are due to defects in mineral deposition.

To monitor the effect of ROCK inhibition on mineral deposition rate and distribution, we applied two pulses of different calcein dyes: blue calcein was added at 29hpf and washed after one hour, followed by green calcein added at 44hpf and washed after one hour. Hence, blue calcein marks the initial triradiate region and green calcein marks the edges of the spicules that were recently deposited (blue and green arrows in **Fig. 3H, I**). The dye FM4-64 was used to mark the cell membranes. Under ROCK inhibition, the green labeled region is much closer to the blue labeled region compared to control, due to the reduction in skeletal elongation rate, in agreement with our SR-μCT measurements. However, while in control embryos each body rod has a single tip stained in green calcein, under ROCK inhibition one of the body rods has two tips and both are stained in green. This indicates that mineral is being deposited in both edges, and the mechanism that prevents the growth of multiple tips in each rod, enabling “tip-dominance”, (in analogy to plant stem apical dominance), is disrupted under ROCK inhibition.

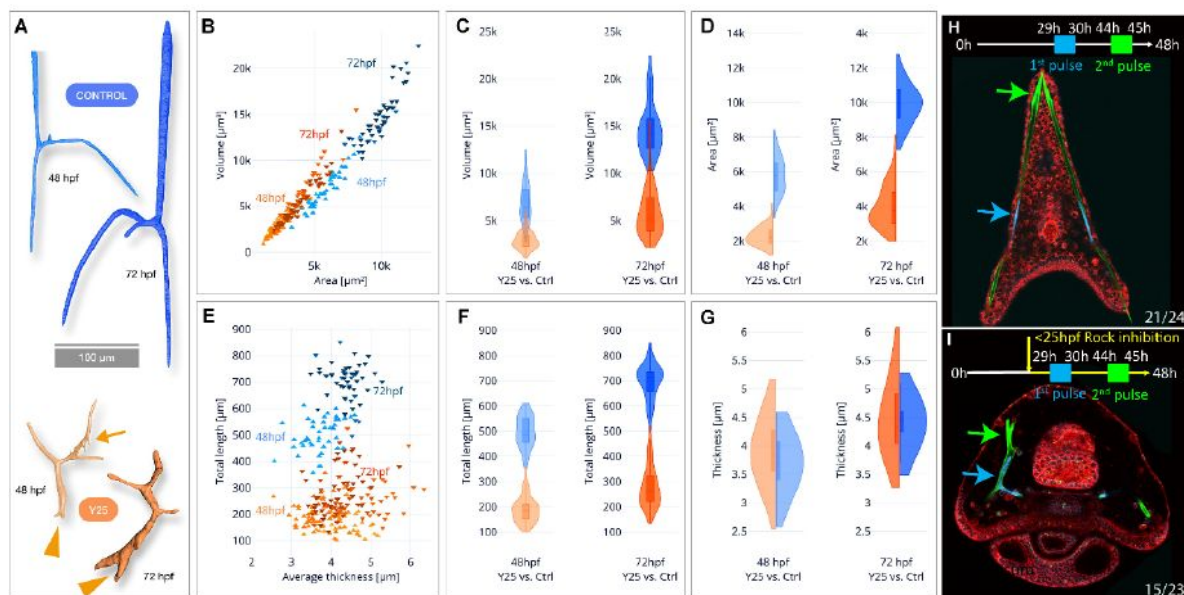


Figure 3

SR-μCT studies of ROCK inhibited spicuels show reduction in skeletal volume, surface area and total length, but not thickness and 2-pulse calcein shows loss of tip-dominance

A, Exemplary 3D-renderings of control spicules (top, blue) and spicules where 40μM of ROCK inhibitor was added at 25hpf (bottom, orange), dissected at 48hpf and 72hpf. Arrowheads point to tip splitting and arrow at 48hpf point to spicule dripping at the back. B, Spicule volume vs. area for control and ROCK-inhibited spicules at 48 and 72 hpf. Each data point represents a single spicule. C-D, Frequency distributions for volume and surface area of control and ROCK-inhibited spicules at 48 hpf and 72 hpf (left and right violin plots, respectively). E, Spicules' total branch length and average thickness for control and ROCK-inhibited spicules at 48 and 72 hpf. F-G, Frequency distributions for spicule lengths and thickness of control and ROCK-inhibited spicules dissected at 48hpf and 72hpf. H, I, Calcein two-pulses experiment. Embryos were exposed to blue calcein from 29-30hpf, to green calcein from 44-45hpf and stained with FM4-64 membrane marker (red) prior to image acquisition at 48hpf. H, Control embryo, G, Embryo where 30μM of Y27632 was added at 25hpf. The experiments were done in three biological replicates and the numbers at the bottom indicate the number of embryos that show this phenotype out of all embryos scored.

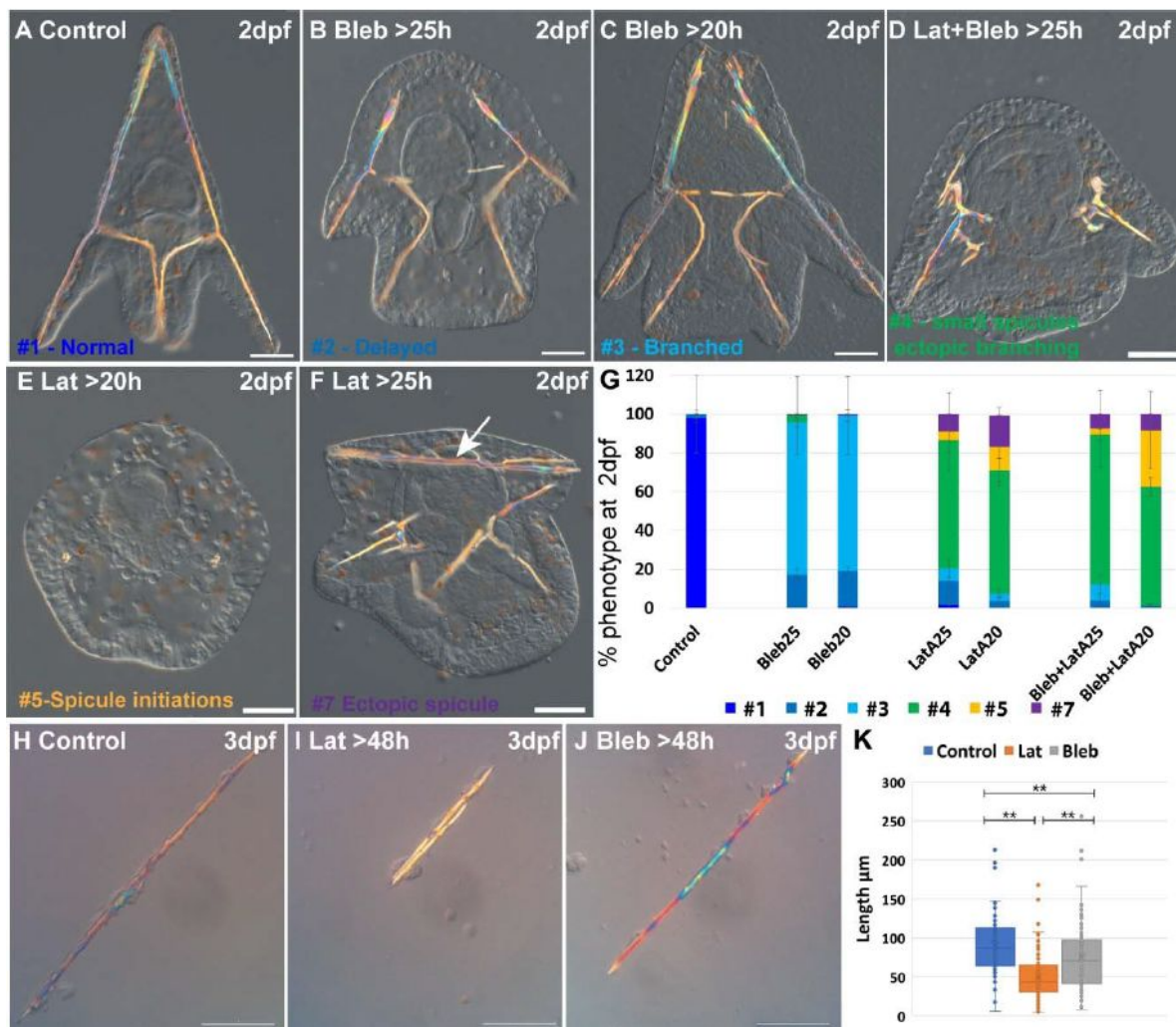


Figure 4

Actin polymerization and myosin activation affect skeletal growth and branching

A-F, representative embryos showing the effect of actomyosin perturbations at 2dpf. A, control embryo B, embryo treated with skeletogenic cell culture were, C, embryo treated with 2µM Blebbistatin >20hpf, D, embryo treated with 2nM Latrunculin-A and 1.5µM Blebbistatin after 25hpf, E, embryo treated with 2nM Latrunculin-A after 20hpf, F, embryo treated with 2nM Latrunculin-A after 25hpf, arrow pointing to the additional spicule rod. G, Statistics of Latrunculin-A and Blebbistatin phenotypes, color code of phenotype is indicated in the representative images. Error bars indicate standard deviation. All treatments were conducted in at least three biological replicates and exact number of replicates and scored embryos are provided in table S4. H-J, representative spicules recorded at 72hpf from H, control skeletogenic cell culture, I, skeletogenic cell culture were 2nM Latrunculin-A was added at 48hpf and J, skeletogenic cell culture were 2µM Blebbistatin was added at 48hpf. K, Quantification of spicule length in the different treatments at 72hpf **P < 0.001, Kruskal-Wallis non-parametric test. Results are based on three biological repeats for each treatment. Scale bars are 50µm.

The activity of the actomyosin network is essential for normal spicule elongation and inhibition of ectopic branching

In other systems, major roles of ROCK are to control F-actin polymerization and MyoII activation (44–48), hence we wanted to directly test the role of these actomyosin components in sea urchin skeletogenesis. To directly inhibit F-actin polymerization we used Latrunculin-A (Lat-A), that prevents the polymerization of actin monomers (49) and was shown to be effective in sea urchin embryos (50). To directly inhibit actomyosin contractility we used Blebbistatin (Blebb), that interferes with MyoII ATPase activity, prevents actin-myosin interactions (51) and was shown to be effective in sea urchin embryos (52). To prevent interference with the early cell divisions where the actomyosin network plays a critical role (53–55), we added the inhibitors, individually or together, before or after spicule formation (20hpf and 25hpf, see methods for details). We tested the resulting phenotypes at 2dpf (Fig. 4A–G).

Our results indicate that F-actin polymerization and to a lesser extent, actomyosin contractility, are essential for normal skeletal growth and inhibition of ectopic branching. The majority of the embryos where MyoII activation was inhibited show mild ectopic branching mostly at the tips of the rods, and a minor delay in skeletal growth (Fig. 4B, C, G). The majority of the embryos where F-actin formation was inhibited show significant reduction of skeletal growth and severe ectopic skeletal branching (Fig. 4E, F and similar phenotypes to Fig. 4D, statistics in Fig. 4G). In some embryos, additional spicules form (Fig. 4F, G). The co-inhibition of F-actin polymerization and MyoII activity results in skeletogenic phenotypes that resemble, but are slightly stronger, than those of the inhibition of F-actin alone (Fig. 4D, G).

To test whether actin polymerization and myosin activity specifically in the skeletogenic cells underlies the observed skeletogenic phenotypes, we inhibit both in an isolated skeletogenic cell culture (Fig. 4H–K). We added the inhibitors after the spicule formed (>48hpf) and observed the phenotypes at 3dpf. Both inhibitors reduced skeletal growth, with Latrunculin-A having a significantly stronger effect than Blebbistatin (Fig. 4K), in agreement with the effect of these inhibitors in whole embryos. However, differently than in whole embryos, skeletal branching in both Latrunculin-A and Blebbistatin is not affected and is similar to the observed branching in control cultures. The absence of the branching phenotype in the skeletogenic cell culture could be due to the increased rigidity of the substrate that could compensate for the reduce actomyosin activity. Another option is that the branching phenotype in whole embryos could be due to the reduction of the activity of the actomyosin network in non-skeletogenic cells. Nevertheless, these results indicate that normal skeletal growth depends on F-actin polymerization and myosin contractility in the skeletogenic cells.

ROCK activity is required for F-actin organization around the forming spicule

Next, we tested whether ROCK regulates the actomyosin structure and activity during skeletal formation. To accomplish this we used phalloidin, a fluorophore that binds F-actin specifically, an antibody that detects phosphorylated MyoII (MyoIIP, (34)) and the skeletogenic marker, 6a9 (Fig. 5). In control embryos, F-actin is detected around the tri-radiate calcite spicules at 27hpf (green arrow in Fig. 5B, in agreement with (34, 35)). The pseudopodia cable that connects the skeletogenic cells shows lower F-actin signal compared to the signal around the spicule (blue line in Fig. 5B, C, E blue arrows in Fig. S7B, C, E). MyoIIP signal is not enriched in the skeletogenic cells nor in the spicule (Fig. 5D, S7D). Under continuous ROCK inhibition, the pseudopodia that connects the skeletogenic cells still forms (Fig. S7H) and F-actin accumulation is still observed in the lateral skeletogenic cell clusters; but unlike control embryos, F-actin is not organized and the spicule cavity is not formed (Fig. 5G). MyoIIP signal seems unchanged by

ROCK inhibition (**Fig. 5I**, S7I), in agreement with the weak and late phenotype of MyoII inhibition (**Figs. 4**). These findings suggest that ROCK activity is essential for F-actin organization around the spicules, but not for the formation of the pseudopodia cable.

F-actin is enriched at the tips of the elongating spicules independently of ROCK activity

To assess the role of ROCK in actomyosin organization during skeletal elongation we compared F-actin and myoIIP signals between control embryos and embryos where ROCK inhibitor was added at 25hpf (**Fig. 5K-T**, S7). At the prism stage, F-actin is enriched at the tips of the spicules in both control and ROCK inhibited embryos (white arrowheads in **Figs. 5L, O**). We quantified the F-actin signal at the tips compared to the back in both treatments (Fig. S7K, L, see methods for details). The phalloidin signal is on average 3-fold stronger at the tips of the spicules compared to the back, in both control and ROCK inhibition, indicating that F-actin enrichment at the tips is significant and independent of ROCK activity. In both control and ROCK inhibited embryos, the F-actin signal was markedly higher in the regions of the pseudopodia cable where the spicule cavity had formed, compared to the regions where the spicule cavity had not yet formed (blue arrows in **Fig. 5L-O, Q-T**). We did not detect an enhanced signal of MyoIIP around the spicules or in the skeletogenic cells, in either condition (**Fig. 5N, S**). At the pluteus stage, F-actin is enriched at the tips of the spicules in both control and ROCK inhibited embryos (white arrowheads in Fig. S7N-Q, S-V). Some non-skeletogenic cells that are enriched with MyoIIP signal are detected at this time (red arrowheads in Fig. S7N-P). Together these data demonstrate that F-actin coats the spicule cavity and is enriched at the tip of the rods, independently of ROCK activity.

We tested the effect of ROCK inhibition on F-actin in skeletogenic cell cultures. F-actin is detected at the tips, in both control cultures and cultures where ROCK was inhibited after spicule formation (**Fig. 5U, V**). Yet, under ROCK inhibition, branching is enhanced and F-actin is enriched at the splitting tips of the spicule rods (Fig. 5V). The enrichment of F-actin at the splitting tips of the spicule under ROCK inhibition in skeletogenic cell culture resembles the calcein staining at the two tips in the embryo in this condition (**Fig. 3I**). These observation further supports the role of ROCK activity in regulating tip-dominance during sea urchin skeletogenesis.

ROCK activity is essential for normal skeletogenic gene expression

The role of ROCK in regulating gene expression during vertebrates' chondrocytes, osteoblasts and odontoblasts differentiation (20–22) intrigued us to study the role of ROCK in regulating skeletogenic gene expression in the sea urchin embryo. ROCK continuous inhibition significantly downregulates the expression of multiple skeletogenic genes, including the cytoskeleton remodeling gene, *rhogap24l/2*, the biomineralization genes, *caral7*, *scl26a5* and *SM30* at 27hpf and 2dpf (**Fig. 6A**, (6)). ROCK inhibition did not affect the expression level of VEGFR and ROCK itself at these times (**Fig. 6A**). Washing the inhibitor after 25hpf or adding the inhibitor at 25hpf had a weaker effect, but still led to downregulation of *caral7*, *SM30*, *angio1* and *notch1* (**Fig. 6B**). Thus, ROCK activity is required for the normal expression level of multiple skeletogenic genes during sea urchin skeletogenesis.

Both continuous and late ROCK inhibition, strongly affect the spatial expression of key regulatory and biomineralization genes that are normally expressed differentially within the skeletogenic lineage at the pluteus stage (**Figs. 6D-F**, S8). These include, the VEGF receptor (VEGFR), the biomineralization genes *SM50* and *SM30*, skeletogenic transcription factors, *Ets1*, *Alx1* and *Erg1* that are essential for skeletogenic cell specification and biomineralization (23, 56) and the transcription factors *Hex* and *MyoD1*. In normal plutei these genes are differentially expressed within skeletogenic lineage, but in both continuous and late ROCK inhibition the expression of these genes becomes localized to the two lateral skeletogenic cell clusters and their vicinity (**Fig. 6D-F**, S8). The lateral skeletogenic cells clusters are the cells where these genes are expressed in

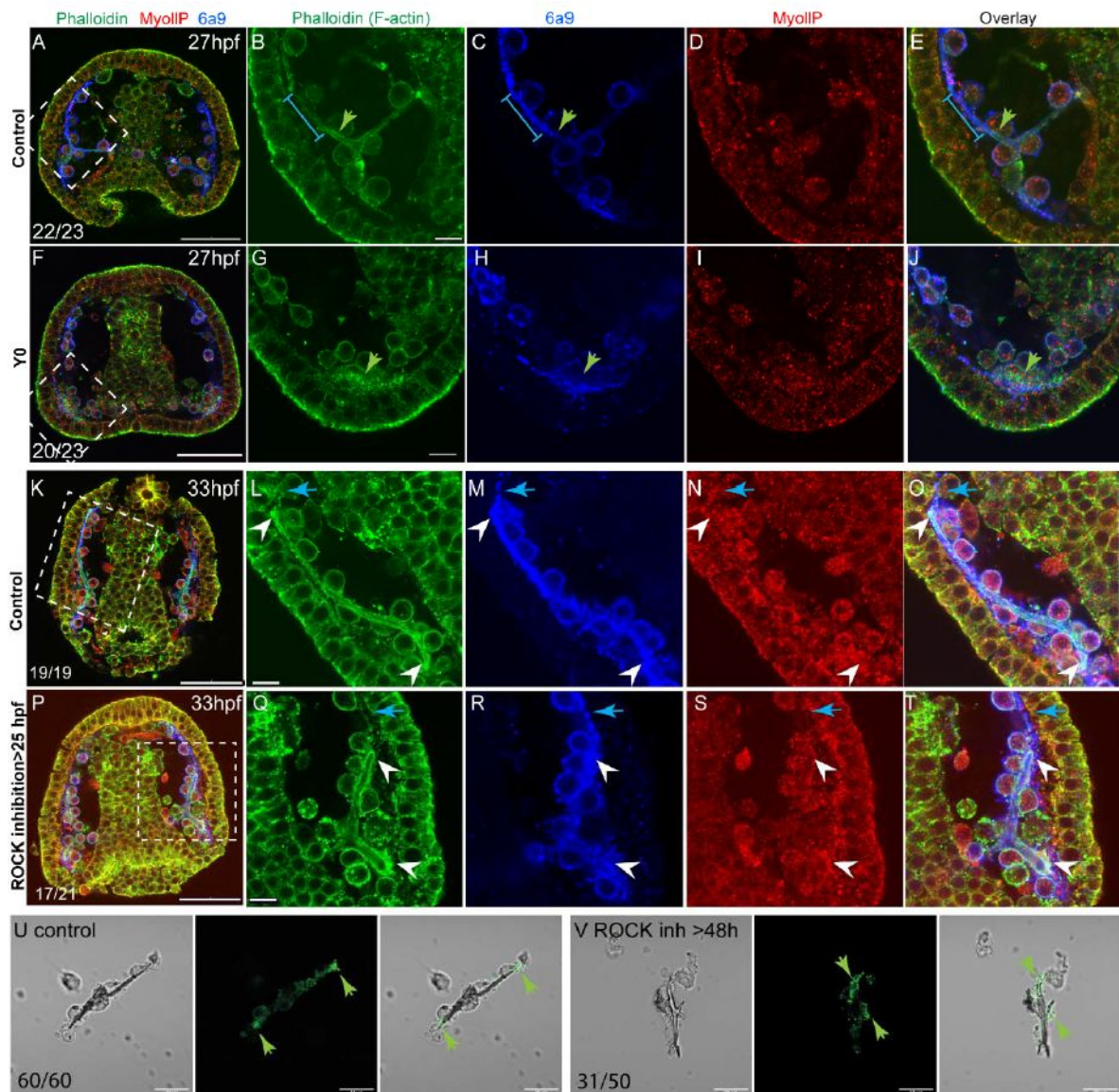


Figure 5

ROCK inhibition effect on F-actin organization and MyoII activity

A-T, Representative images showing normal embryos and embryos treated with 80μM ROCK inhibitor at different times. Right panels show enlargements of the rectangle sections marked in the left most panels. Phalloidin (green) marks F-actin, MyoIIP (red) marks phosphorylated MyoII and 6a9 (blue) marks the skeletogenic cells. A-E, Representative images at 27hpf showing control embryo and F-J, embryos treated with ROCK inhibitor from fertilization. Green arrows point to regions that show enriched F-actin signal. K-T, representative images at 33hpf, showing normal embryo K-O, and P-T, embryos where 30μM ROCK inhibitor was added at 25hpf. White arrowheads point to the enriched F-actin signal at the tips. Blue arrows point to the region of the pseudopodia cable that is not filled with the spicule cavity. The experiments were done in 3 biological replicates, the numbers at the bottom left of (A, F, K, P) indicate the number of embryos that show this phenotype out of all embryos scored. Scale bar in A, F, K, P is 50μm, and in B, G, L, Q is 10μm. U, V, representative spicules out of three biological replicates from control skeletogenic cell culture (n=60), U, and skeletogenic cell cultures that was treated with 30μM ROCK inhibitor was added at 48hpf (n=52), recorded at 72hpf. Left panel is phase image, middle panel is phalloidin staining and right panel shows the overlay. Green arrows point to the enhanced F-actin signal at the tips. Scale bar is 20μm.

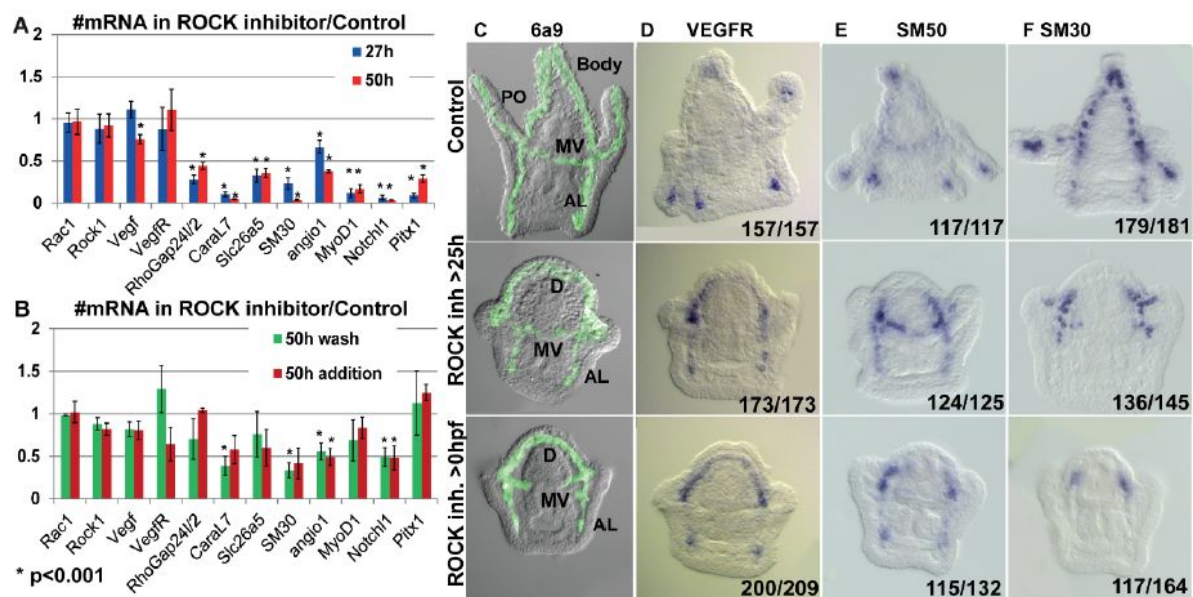


Figure 6

ROCK activity is essential for normal gene expression in the skeletogenic cells

A, B, the effect of 80μM ROCK inhibition on gene expression. A, continuous ROCK inhibition at 27hpf and 50 hpf (n=4). B, addition of ROCK inhibitor at 25hpf and the wash of ROCK inhibitor at 25hpf, measured at 50hpf (n=3). Asterisks indicate $p < 0.001$, one tailed z-test. Error bars show standard deviation. C-F, Representative images of control embryo (top panels), embryos where ROCK inhibitor was added at 25hpf (middle panels), and embryos that were exposed to continuous ROCK inhibition (bottom panels), at the pluteus stage (~48hpf). C, skeletogenic cell marker, 6a9. MV, midventral; AL, anterolateral, and PO, Post-oral rods. D, dorsal skeletogenic chain. D-F, spatial expression of skeletogenic genes. Gene names are indicated at the top of each panel. Numbers at the bottom of each image indicate the number of embryos that show this phenotype (left) out of all embryos scored (right), conducted in at least three independent biological replicates.

the gastrula stage (24-27hpf) and where spicule formation begins in normal embryos (6, 32). The expression of VEGFR and SM50 under ROCK inhibition is more expanded than the other tested genes and is also observed at the anterolateral rods (Fig. 6B, C). The VEGF ligand that is normally expressed in the ectodermal cells near the four tips of the post-oral and anterolateral rods (32), is expressed at four ectodermal domains near the lateral skeletogenic cell clusters and near the end of the anterolateral chain, under ROCK inhibition (Fig. S8G).

Overall, under ROCK inhibition, the expression of key regulatory and biomineralization genes is downregulated (Fig. 6A) and remains localized to the area near the lateral skeletogenic cells clusters (Fig. 6D-F, S8). This localization is despite the proper formation of the dorsal, anterolateral and mid-ventral chains (Fig. 6C). This implies that ROCK activity provides regulatory cues to the skeletogenic GRN required for the dynamic progression of gene expression within the skeletogenic lineage.

Discussion

Biomineralization is a complex morphogenetic process, encoded in the species genome and executed by the GRNs and the cellular machinery. Here we used the sea urchin larval skeletogenesis to investigate the role of ROCK and the actomyosin network in biomineral formation and in regulating gene expression, as we summarize in Fig. 7. ROCK expression is enriched in the skeletogenic cells, partially depending on VEGF signaling (Fig. 1). ROCK activity in the skeletogenic cells is necessary for spicule formation, skeletal elongation and prevention of ectopic skeletal branching (Figs. 2-3, S3, 4). Direct inhibition of F-actin polymerization results in similar but not identical skeletogenic phenotypes (Fig. 4) and F-actin is organized around the spicule cavity and enriched at the spicule tips (Figs. 5, S7). ROCK activity feeds back into the skeletogenic GRN and affects regulatory and biomineralization gene expression (Fig. 6). Below we discuss our findings and their implications for the biological control and evolution of biomineralization.

The first step in generating the sea urchin spicules is the construction of the spicule cavity, a membrane filled with calcium carbonate and coated with F-actin (Fig. 7A). ROCK activity is necessary for the formation of the spicule cavity and for the organization of F-actin around it (Figs. 2, 5). A direct perturbation of F-actin polymerization strongly disrupts spicule growth and shape (Fig. 4). F-actin is enriched in the regions of the pseudopodia cable that are filled with mineral, supporting its role in mineral deposition (Figs. 5, S7, Fig. 7A). Overall, our studies indicate that ROCK activity in the skeletogenic cells is essential for the formation of the spicule cavity and for the initial F-actin organization around it, which could be essential for mineral deposition into it (Fig. 7C).

After the spicule cavity has formed, the spicules elongate inside the pseudopodia chord by mineral deposition at the tips of the spicule rods, which depends on ROCK activity and on F-actin polymerization (Figs. 2-4, S7, 7D-F). F-actin is detected along the spicules and is significantly enriched at the tips of the elongating spicules (Fig. 4, S7 (S7)). F-actin enrichment at the tips is independent on ROCK activity (Fig. 5K-T, S7K, F), yet under ROCK inhibition and under the inhibition of F-actin polymerization, ectopic branching is observed at the tips and in the back (tip splitting and spicule dripping, Figs. 2J, N, 3A, 4, 5V). The participation of F-actin and ROCK in polarized growth and vesicle exocytosis has been observed in both animals and plants (S7, S8). F-actin is used for short range myosin trafficking of vesicles toward the secretion site (S7, S9). Once the secretory vesicle fuses with the membrane, it is coated with F-actin that stabilizes the vesicle and enables MyoII binding and ROCK-regulated contraction, necessary for content secretion (60, 61). In plant cells, F-actin accumulate at the growing tip of the cell and assists vesicle exocytosis necessary for polarized tip-growth (S8). Our observations suggest that ROCK, F-actin and MyoII could be regulating the localized exocytosis of mineral-bearing vesicles

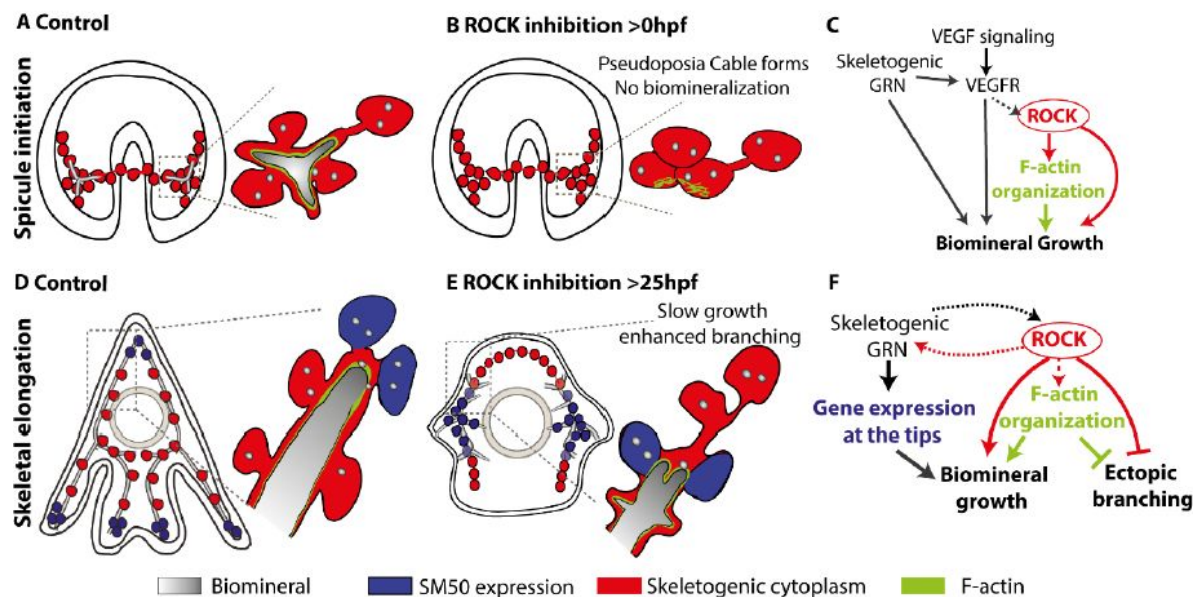


Figure 7

The role of ROCK in sea urchin skeletogenesis, summary

A, at the gastrula stage (~27hpf in *P. lividus*), the triradiate spicule is coated with F-actin. B, ROCK activity is required for spicule formation and F-actin organization. C, model of the regulatory interactions between the skeletogenic GRN and cytoskeleton remodeling during spicule initiation (~24-27hpf in *P. lividus*). D, during skeletal elongation F-actin is detected around the spicule cavity and is enriched at the tips of the rods. The expression of SM50 and some other skeletogenic genes, is localized to the tips of the rods. E, under the addition of ROCK inhibitor, skeletal growth rate is reduced and ectopic spicule branching is observed. The expression of skeletogenic genes is localized to the vicinity of the growing rods. F, the regulatory interactions between the skeletogenic GRN and cytoskeleton remodeling during skeletal elongation.

during sea urchin skeletogenesis. The delay in spicule growth and the ectopic branching could be due to impaired and misplaced vesicle trafficking and deposition under ROCK and F-actin inhibition, but the exact protein regulatory network is still to be determined.

ROCK activity is necessary for correct spatiotemporal expression of regulatory and biomineralization genes (Fig. 6). ROCK inhibition downregulates the expression level of skeletogenic genes (Fig. 6A) and prevents the spatial progression of key skeletogenic genes toward the edges of the skeletal rods (Fig. 7E). In other systems, ROCK activity was shown to regulate gene expression through various intracellular pathways (62–64). Furthermore, ROCK has an important role in mechano-sensing of matrix rigidity and transducing it into changes in gene expression (65). Relatedly, stiff matrix activates ROCK in pre-osteoblastic cells, which activates Erk that leads to osteogenic differentiation and upregulation of osteogenic gene expression (64). Sea urchin ROCK could be sensing the spicule stiffness and transducing it into gene expression, providing a necessary feedback between spicule elongation and the skeletogenic GRN. Further studies are required to identify the specific molecular mechanisms activated by ROCK to affect gene expression in the skeletogenic cells.

Overall, our findings and the role of actomyosin remodeling in biomineralization, from calcifying and silicifying single cells to vertebrates' bones and teeth (15–22, 64), suggest that this molecular machinery is a part of the common molecular tool-kit of biomineralization. Most likely, the actomyosin network was employed independently, downstream of distinct GRNs across Eukaryotes, to control mineral growth and morphogenesis.

Methods

Animal and embryos

Adult *Paracentrotus lividus* were obtained from the Institute of Oceanographic and Limnological Research (IOLR) in Eilat, Israel. Eggs and sperm were obtained by injection of 0.5M KCl solution into the adult sea urchin species. Embryos were cultured at 18°C in 0.2μ filtered ASW.

Imaging

Embryonic phenotypes and WMISH were imaged by Zeiss Axioimager 2. Fluorescent markers were imaged using Nikon A1-R Confocal microscope. All images were aligned in Adobe photoshop and Illustrator.

Axitinib (AG013736) treatment

Axitinib (AG013736, Selleckchem, Houston, TX, USA) was applied as described in (32).

Western Blot

Embryo cultures treated with 150 nM Axitinib, or DMSO as control, were collected at 22hpf, 27hpf and 33hpf by centrifugation of 10,000g for 2 minutes. We lysed the pellet in 150μL lysis buffer (20 mM Tris-HCl, 150 mM NaCl, 1% Triton X-100, pH 8) with protease inhibitors (protease inhibitor cocktail w/o metal chelator; Sigma, P8340; 1:100) and phosphatase inhibitors (50 mM NaF, 1 mM Na3VO4, 1 mM Na4P2O7, 1mM, βGP) as described in (66). 40μg protein were loaded on 8% SDS-acrylamide gel, transferred to PVDF membranes for Western blot analysis. Anti-ROCK antibody (ab45171, Abcam) was used in a 1:300 dilution followed by anti-rabbit HRP secondary antibody diluted to 1:2500 (Jackson ImmunoResearch 111-035-003). For the loading control, the membrane was incubated with anti-β-tubulin antibody (1:5000; Sigma) followed by anti-mouse antibody 1:5000 (Jackson ImmunoResearch 115-035-003). Quantification of ROCK signal was done using Image studio lite vr. 5.2. ROCK signal was divided by Tubulin signal and then the ratio between

ROCK/Tubulin in control vs. VEGFR inhibition was calculated. The graph in Fig. S2 was generated based on three biological replicates (different sets of parents) at 27hpf and 33hpdf and four biological replicates at 22hpf.

Immunostaining procedure

Phalloidin labeling, MyoIIP (p20-MyosinII), ROCK (anti ROCK2+ROCK1 antibody [EP786Y], AB-ab45171, Abcam), and 6a9 immunostaining were done similarly to (34 [↗](#)).

Quantification of Anti-ROCK and Phalloidin signal

A graphical program was written to allow manual quantification of the fluorescent signal from the spicule regions by identifying “stained” pixels per selected area. The ratio of stained anti-ROCK to region total area was compared between ectodermal and skeletogenic cells, and between control and VEGFR inhibited embryos at 22, 27 and 33hpf. The average ratios and ztest for significance difference from 1, are 22hpf control, ~ 1.3 , $z=0.0006$; VEGFR inhibition ~ 1 , $z=0.5$, 27hpf control ~ 1.8 , $z<10^{-11}$; VEGFR inhibition ~ 1.6 , $z<10^{-8}$, 33hpf control ~ 4.6 , $z<10^{-23}$; VEGFR inhibition ~ 1.5 , $z<10^{-5}$. For phalloidin analyses, embryos at 33hpf of control and ROCK inhibitor addition at 25hpf were used and for each image, two regions were selected: an area at the tip of the spicule and an area at the back of the spicule, about ~ 10 microns apart. The ratio of stained phalloidin to region total area was compared between tip and back, in control and ROCK inhibited embryos. Paired t-test was used to compare the differences in phalloidin signal between the tip and back of the spicules across all groups and between control and ROCK inhibited embryos.

ROCK MASOs injections

Microinjections were conducted similarly to (32 [↗](#)). The eggs were injected with an injection solution containing 0.12M KCl, 0.5 μ g/ μ l Rhodamine Dextran and 900 μ M of MASOs. Translation MASOs were designed and synthesized by Gene Tools, Philomath, OR, according to ROCK sequence (39 [↗](#)). The sequence of ROCK MASO-1 (5'-AGACATATTGGAGCCGA[CAT]CCTG-3') matches the start of translation region and ROCK MASO-2 (5'-TCTCTGCGTTATATCCACTAAGT-3') matches the 5' region upstream of the start of translation (39 [↗](#)). Control embryos were injected in the same concentration with Random MASO. Injected embryos were cultured at 18°C, imaged and scored at 2dpf.

ROCK inhibitor (Y27632) treatment

Y27632 stock (10005583 CAS Registry No. 129830-38-2, Cayman chemical), was prepared by reconstituting the chemical in DMSO. The embryos were treated with Y27632 at a final concentration between 30-80 μ M, as mentioned in the results. Throughout the paper, control embryos were cultured in DMSO at the same volume as Y27632 solution and no more than 0.1% (v/v).

ROCK inhibition in isolated skeletogenic cell culture

Skeletogenic cell culture was performed as described in (36 [↗](#)) with minor changes. Isolated skeletogenic cells were cultured in CultureSlides (4 chambers polystyrene vessel tissue culture treated glass slide REF 354104) with ASW (Red Sea Fish LTD) containing Gentamicin and Penicillin-Streptomycin (GPS) at 18°C. Media containing 4% (v/v) horse serum (Sigma-Aldrich H1270) in ASW+GPS was added to the culture. ROCK inhibitor, Y27632, was added to the cell culture at 16hpf (before spicule initiation) or 48hpf (after spicule initiation), and images were taken at 72hpf. Each treatment was conducted in three biological repeats except from the >48hpf addition of 120 μ M which was done in two biological repeats.

Quantification of skeletal length and statistical analysis

Skeletal length measurement was done as described in (32 [DOI](#)). The measurements were repeated for three biological repeats for control, ROCK inhibition with 30 μ M and 80 μ M, Latrunculin-A 2nM, Blebbistatin 2 μ M, and 2 biological repeats for 120 μ M. In ROCK inhibition experiments, a total of 275 skeletons were measured for control, 105 skeletons for 30 μ M, 122 skeletons for 80 μ M and 80 for 120 μ M. In Latrunculin-A and Blebbistatin experiments, a total of 116 skeletons were measured for control, 149 for Latrunculin-A and 107 for Blebbistatin. The data were analyzed in Excel and the statistical analysis was performed using Kruskal-Wallis non-parametric test in Excel as described in (67 [DOI](#)) and SPSS statistics 27.

Spicule preparation for micro-computed tomography (SR- μ CT) measurement

Control embryos and embryos where 40 μ M Y27632 was added at 25hpf, were collected by centrifugation of 50mL Falcon tubes at 2000rpm, at 48h and 72hpf. Every tube was washed in 10 mL cold distilled water (DW), transferred into epi-tube and washed three times in 1mL cold DW (incubation at 4°C between every wash, centrifuge at 2000rpm in room temperature for 2min). Then the skeletons were washed three times in 3% NaOCl followed by three washes in DW, one wash in 70% EtOH and one wash in Acetone. Skeletal spicules were dried at room temperature and stored at -20 °C.

Synchrotron and lab-based SR- μ CT

Dry and loose skeletal spicules prepared as described above, were used for the SR- μ CT analysis. We analyzed 14 sets of spicules from three (control) and four (ROCK inhibited) pairs of parents, divided into four groups: 2dpf, control, n=44; 2dpf ROCK inhibited, n=93; 3dpf control, n=51; 3dpf ROCK inhibited, n=93, see Table S1). Spicules were glued to sharpened toothpick tips, either in bulk or individually when picked by hand by means of a brush bristle. Each toothpick loaded with spicules, was mounted on a 3ml vial twist-off lid and for transportation securely stored in the vial (Fig. S5A, B). For each set 2-3 μ CT-samples were prepared, which allowed the acquisition of statistically relevant sample sizes (40+ spicules/group, Table S1). Tomographic data sets were acquired using the following scanning parameters; for the lab based μ CT (RX Solutions, Chavanod, France) 90kV, 143-160 μ A, 1440 images, average frames 10, frame rates 0.4-1, voxel sizes: 0.45-59 μ m; and for the synchrotron radiation μ CT (Anatomix, Synchrotron SOLEIL, Gif-sur-Yvette, France) 2048 images, angle between projections 0.09°, voxel size: 0.65 μ m. Tomographic data reconstruction was performed with commercial (RX Solutions, Chavanod, France) and PyHST2 software (A. Mirone et al., Nucl. Instrum. Meth. B 324 (2014) 41-48, doi:10.1016/j.nimb.2013.09.030) at the MPICI and at Anatomix, SOLEIL, respectively.

Three-dimensional (3D) data analysis

Data visualization, pre-processing including cropping and image type conversion to 16bit (of the Sr- μ CT data), and user-augmented segmentation of intact spicules were performed in Amira using off-the-shelf modules (Fig. S5C-E, (68 [DOI](#))). Volume and area measurements were performed on the segmented spicules, i.e. the label-fields using Amira's material.statistics module (Volume3D and Area3D). Average spicule thickness and total length measurements were performed with python code, utilizing available python libraries (e.g., skimage, scan, scipy). The code was applied to 3D-tif files created from Amira's label-fields, which contained the segmented spicules. For length measurements, the spicules were skeletonized and the chord lengths of the branches forming a spicule-skeleton were summed up to obtain the total spicule length. For thickness measurements, we performed Euclidean distance transforms for each spicule using scipy's ndimage.morphology.distance_transform_edt package, and calculated the average spicule thickness based on the distance measurements obtained at each voxel from spicule-skeleton to the nearest spicule surface. Quality control of our measurements was performed with open source software, Fiji and available plugins (e.g. BoneJ, FiberJ, DiameterJ).

Calcein staining

A 2mg/ml stock solution of blue calcein (M1255, Sigma) and a 2mg/ml stock solution of green calcein (C0875, Sigma) were prepared by dissolving the chemicals in distilled water. Working solution of 250µg/ml and 150µg/ml, respectively, was prepared by diluting the stock solution in artificial sea water. Blue calcein was added to the embryo culture at 29hpf for one hour and then washed. At 44hpf the green calcein was added for one hour and washed. For FM4-64 staining A 100µg/ml stock solution of FM4-64 (T13320, Life technologies, OR, USA) was prepared by dissolving the chemical in distilled water. Working solution of 2.5 µg/ml was prepared by diluting the stock solution in artificial sea water. Embryos were immersed in working solution about 10 minutes before visualization.

Latrunculin-A and Blebbistatin treatments

1mM of Lat-A stock (L12370, Thermo Fisher Scientific) was prepared by reconstituting the chemical in DMSO. The embryos were treated with Lat-A at final concentration of 2 nM. 1mM of Blebbistatin stock (13013-5, Enco) was prepared by reconstituting the chemical in DMSO. The embryos were treated with Blebbistatin at final concentration of 2 µM. For the double inhibition experiments, embryos were treated with Blebb at final concentration of 1.5 µM, and with Lat-A at final concentration of 2 nM.

WMISH procedure

WMISH was performed as described in [\(32\)](#). List of the primers used to make the WMISH probes is available in [\(32\)](#).

cDNA preparation for QPCR experiments

For ROCK inhibition experiments total RNA was extracted from >1000 sea urchin embryos in each condition using RNeasy Mini Kit [\(50\)](#) from QIAGEN (#74104) according to the kits' protocol. DNase treatment on column was done using RNase-Free DNase Set-Qiagen [\(50\)](#) (#79254). RNA was reverse-transcribed using the High Capacity cDNA RT kit, AB-4368814 (Applied Biosystems) following the manufacturer's instructions.

Quantitative polymerase chain reaction (QPCR)

QPCR was carried out as described in [\(32\)](#). Complete list of primer sequences used for the experiments is provided there.

Acknowledgements

We are grateful to SOLEIL for provision of synchrotron radiation facilities and we would like to thank Dr. Mario Scheel, Dr. Timm Weitkamp and Dr. Jonathan Perrin for assistance in using beamline ANATOMIX for nano and µCT. We thank the Bioimaging Unit, Faculty of Natural Sciences, University of Haifa for assistance with the use of confocal. We thank Charles Etensohn for the generous gift of the 6a9 antibody produced in his lab. We thank David Ben-Ezra and Michael Kantorovitz for their help with sea urchin handling. We thank Haguy Wolfenson for illuminating discussions of the results and conceptual framework. We thank Nir Sapir and Tovah Nehrer for their advice and help with the statistical analysis of spicule length. We thank Alvaro Israel and Omri Nahor for their help with sea urchin maintenance. We thank Yulia Kagan for her help with the western blot.

Funding

Israel Science Foundation grant number 211/20 (S.B.D.), and Zuckerman fellowship (M.R.W.), TUD internal funding (YP). ANATOMIX is an Equipment of Excellence (EQUIPEX) funded by the Investments for the Future program of the French National Research Agency (ANR), project NanoimagesX, grant no. ANR-11-EQPX-0031. Access to ANATOMIX beamline is granted for proposal number proposal 20200811.

References

1. Weiner S, Addadi I (2011) **Crystallization Pathways in Biomineralization** *Annu Rev Mater Res* **41**:21–40
2. Murdock DJ, Donoghue PC (2011) **Evolutionary origins of animal skeletal biomineralization** *Cells Tissues Organs* **194**:98–102
3. Knoll AH (2003) **Biomineralization and Evolutionary History** *Reviews in Mineralogy and Geochemistry* **54**:329–56
4. Gilbert P, Bergmann KD, Boekelheide N, Tambutte S, Mass T, Marin F, et al. (2022) **Biomineralization: Integrating mechanism and evolutionary history** *Sci Adv* **8**
5. Ben-Tabou de-Leon S (2022) **The Evolution of Biomineralization through the Co-Option of Organic Scaffold Forming Networks** *Cells* **11**
6. Morgulis M, Gildor T, Roopin M, Sher N, Malik A, Lalzar M, et al. (2019) **Possible cooption of a VEGF-driven tubulogenesis program for biomineralization in echinoderms** *Proc Natl Acad Sci U S A* **116**:12353–62
7. Murdock DJE (2020) **The ‘biomineralization toolkit’ and the origin of animal skeletons** *Biol Rev Camb Philos Soc* **95**:1372–92
8. Hojo H, McMahon AP, Ohba S (2016) **An Emerging Regulatory Landscape for Skeletal Development** *Trends Genet* **32**:774–87
9. Fraser GJ, Hulsey CD, Bloomquist RF, Uyesugi K, Manley NR, Streelman JT (2009) **An ancient gene network is co-opted for teeth on old and new jaws** *PLoS Biol* **7**
10. Evans JS (2019) **The Biomineralization Proteome: Protein Complexity for a Complex Bioceramic Assembly Process** *Proteomics* **19**
11. Khor JM, Ettensohn CA (2017) **Functional divergence of paralogous transcription factors supported the evolution of biomineralization in echinoderms** *eLife* **6**
12. Kawasaki K, Buchanan AV, Weiss KM (2009) **Biomineralization in humans: making the hard choices in life** *Annu Rev Genet* **43**:119–42
13. Drake JL, Mass T, Haramaty L, Zelzion E, Bhattacharya D, Falkowski PG (2013) **Proteomic analysis of skeletal organic matrix from the stony coral *Stylophora pistillata*** *Proc Natl Acad Sci U S A* **110**:3788–93
14. Tucker A, Sharpe P (2004) **The cutting-edge of mammalian development; how the embryo makes teeth** *Nat Rev Genet* **5**:499–508
15. Langer G, de Nooijer L Jan, Oetjen K (2010) **On the role of cytoskeleton in coccolith morphogenesis: the effect of cytoskeleton inhibitors** *Journal of Phycology* **46**:1252–6
16. Durak GM, Brownlee C, Wheeler GL (2017) **The role of the cytoskeleton in biomineralisation in haptophyte algae** *Scientific reports* **7**

17. Tyszkaj J, Bickmeyer U, Raitzsch M, Bijma J, Kaczmarek K, Mewes A, et al. (2019) **Form and function of F-actin during biomineralization revealed from live experiments on foraminifera** *Proc Natl Acad Sci U S A* **116**:4111–6
18. Tesson B, Hildebrand M (2010) **Dynamics of silica cell wall morphogenesis in the diatom *Cyclotella cryptica*: substructure formation and the role of microfilaments** *J Struct Biol* **169**:62–74
19. Tesson B, Hildebrand M (2010) **Extensive and intimate association of the cytoskeleton with forming silica in diatoms: control over patterning on the meso- and micro-scale** *PLoS One* **5**
20. Strzelecka-Kiliszek A, Mebarek S, Roszkowska M, Buchet R, Magne D, Pikula S (2017) **Functions of Rho family of small GTPases and Rho-associated coiled-coil kinases in bone cells during differentiation and mineralization** *Biochim Biophys Acta Gen Subj* **1861**:1009–23
21. Qin L, Liu W, Cao H, Xiao G (2020) **Molecular mechanosensors in osteocytes** *Bone Res* **8**
22. Huang X, Chen X, Chen H, Xu D, Lin C, Peng B (2018) **Rho/Rho-associated protein kinase signaling pathway-mediated downregulation of runt-related transcription factor 2 expression promotes the differentiation of dental pulp stem cells into odontoblasts** *Exp Ther Med* **15**:4457–64
23. Oliveri P, Tu Q, Davidson EH (2008) **Global regulatory logic for specification of an embryonic cell lineage** *Proc Natl Acad Sci U S A* **105**:5955–62
24. Etensohn CA, Dey D (2017) **KirrelL, a member of the Ig-domain superfamily of adhesion proteins, is essential for fusion of primary mesenchyme cells in the sea urchin embryo** *Dev Biol* **421**:258–70
25. Gildor T, Winter MR, Layous M, Hijaze E, Ben-Tabou de-Leon S (2021) **The biological regulation of sea urchin larval skeletogenesis - From genes to biomineralized tissue** *J Struct Biol* **107797**
26. Kahil K, Varsano N, Sorrentino A, Pereiro E, Rez P, Weiner S, et al. (2020) **Cellular pathways of calcium transport and concentration toward mineral formation in sea urchin larvae** *Proc Natl Acad Sci U S A* **117**:30957–65
27. Kahil K, Kaplan-Ashiri I, Wolf SG, Rechav K, Weiner S, Addadi L (2023) **Elemental compositions of sea urchin larval cell vesicles evaluated by cryo-STEM-EDS and cryo-SEM-EDS** *Acta Biomater* **155**:482–90
28. Vidavsky N, Addadi S, Mahamid J, Shimon E, Ben-Ezra D, Shpigel M, et al. (2014) **Initial stages of calcium uptake and mineral deposition in sea urchin embryos** *Proc Natl Acad Sci U S A* **111**:39–44
29. Wilt FH, Killian CE, Hamilton P, Croker L (2008) **The dynamics of secretion during sea urchin embryonic skeleton formation** *Exp Cell Res* **314**:1744–52
30. Adomako-Ankomah A, Etensohn CA (2013) **Growth factor-mediated mesodermal cell guidance and skeletogenesis during sea urchin gastrulation** *Development* **140**:4214–25
31. Sun Z, Etensohn CA (2014) **Signal-dependent regulation of the sea urchin skeletogenic gene regulatory network** *Gene Expr Patterns* **16**:93–103

32. Tarsis K, Gildor T, Morgulis M, Ben-Tabou de-Leon S (2022) **Distinct regulatory states control the elongation of individual skeletal rods in the sea urchin embryo** *Dev Dyn*
33. Morgulis M, Winter MR, Shternhell L, Gildor T, Ben-Tabou de-Leon S (2021) **VEGF signaling activates the matrix metalloproteinases, MmpL7 and MmpL5 at the sites of active skeletal growth and MmpL7 regulates skeletal elongation** *Dev Biol* **473**:80–9
34. Winter MR, Morgulis M, Gildor T, Cohen AR, Ben-Tabou de-Leon S (2021) **Calcium-vesicles perform active diffusion in the sea urchin embryo during larval biomineralization** *PLoS Comput Biol* **17**
35. Stepicheva NA, Dumas M, Kobi P, Donaldson JG, Song JL (2017) **The small GTPase Arf6 regulates sea urchin morphogenesis** *Differentiation* **95**:31–43
36. Moreno B, DiCorato A, Park A, Mobilia Kellen, Knapp R, Bleher R, et al. (2019) **Culture of and experiments with sea urchin embryo primary mesenchyme cells** *Methods in Cell Biology* **150**:293–330
37. Sepulveda-Ramirez SP, Toledo-Jacobo L, Henson JH, Shuster CB (2018) **Cdc42 controls primary mesenchyme cell morphogenesis in the sea urchin embryo** *Dev Biol* **437**:140–51
38. Croce J, Duloquin L, Lhomond G, McClay DR, Gache C (2006) **Frizzled5/8 is required in secondary mesenchyme cells to initiate archenteron invagination during sea urchin development** *Development* **133**:547–57
39. Marletaz F, Couloux A, Poulain J, Labadie K, Da Silva C, Mangenot S, et al. (2023) **Analysis of the P. lividus sea urchin genome highlights contrasting trends of genomic and regulatory evolution in deuterostomes** *Cell Genom* **3**
40. Arshinoff BI, Cary GA, Karimi K, Foley S, Agalakov S, Delgado F, et al. (2022) **Echinobase: leveraging an extant model organism database to build a knowledgebase supporting research on the genomics and biology of echinoderms** *Nucleic Acids Res* **50**:D970–D9
41. Ettensohn CA, McClay DR (1988) **Cell lineage conversion in the sea urchin embryo** *Dev Biol* **125**:396–409
42. Chen W, Mao K, Liu Z, Dinh-Xuan AT (2014) **The role of the RhoA/Rho kinase pathway in angiogenesis and its potential value in prostate cancer (Review)** *Oncol Lett* **8**:1907–11
43. Jacobs M, Hayakawa K, Swenson L, Bellon S, Fleming M, Taslimi P, et al. (2006) **The structure of dimeric ROCK I reveals the mechanism for ligand selectivity** *J Biol Chem* **281**:260–8
44. Liu J, Wada Y, Katsura M, Tozawa H, Erwin N, Kapron CM, et al. (2018) **Rho-Associated Coiled-Coil Kinase (ROCK) in Molecular Regulation of Angiogenesis** *Theranostics* **8**:6053–69
45. Julian L, Olson MF (2014) **Rho-associated coiled-coil containing kinases (ROCK): structure, regulation, and functions** *Small GTPases* **5**
46. Lecuit T, Lenne PF, Munro E (2011) **Force generation, transmission, and integration during cell and tissue morphogenesis** *Annu Rev Cell Dev Biol* **27**:157–84
47. Amin E, Dubey BN, Zhang SC, Gremer L, Dvorsky R, Moll JM, et al. (2013) **Rho-kinase: regulation, (dys)function, and inhibition** *Biol Chem* **394**:1399–410

48. Pavlov D, Muhlrads A, Cooper J, Wear M, Reisler E (2007) **Actin filament severing by cofilin** *J Mol Biol* **365**:1350–8
49. Yarmola EG, Somasundaram T, Boring TA, Spector I, Bubb MR. (2000) **Actin-latrunculin A structure and function. Differential modulation of actin-binding protein function by latrunculin A** *J Biol Chem* **275**:28120–7
50. Schatten G, Schatten H, Spector I, Cline C, Paweletz N, Simerly C, et al. (1986) **Latrunculin inhibits the microfilament-mediated processes during fertilization, cleavage and early development in sea urchins and mice** *Exp Cell Res* **166**:191–208
51. Kovacs M, Toth J, Hetenyi C, Malnasi-Csizmadia A, Sellers JR (2004) **Mechanism of blebbistatin inhibition of myosin II** *J Biol Chem* **279**:35557–63
52. Moorhouse KS, Gudejko HF, McDougall A, Burgess DR (2015) **Influence of cell polarity on early development of the sea urchin embryo** *Dev Dyn* **244**:1469–84
53. Mabuchi I (1994) **Cleavage furrow: timing of emergence of contractile ring actin filaments and establishment of the contractile ring by filament bundling in sea urchin eggs** *J Cell Sci* **107**:1853–62
54. Schroeder TE (1987) **Fourth cleavage of sea urchin blastomeres: microtubule patterns and myosin localization in equal and unequal cell divisions** *Dev Biol* **124**:9–22
55. Wong GK, Allen PG, Begg DA (1997) **Dynamics of filamentous actin organization in the sea urchin egg cortex during early cleavage divisions: implications for the mechanism of cytokinesis** *Cell motility and the cytoskeleton* **36**:30–42
56. Khor JM, Ettensohn CA (2020) **Transcription Factors of the Alx Family: Evolutionarily Conserved Regulators of Deuterostome Skeletogenesis** *Frontiers in genetics* **11**
57. Miklavc P, Frick M (2020) **Actin and Myosin in Non-Neuronal Exocytosis** *Cells* **9**
58. Bibeau JP, Kingsley JL, Furt F, Tuzel E, Vidali L (2018) **F-Actin Mediated Focusing of Vesicles at the Cell Tip Is Essential for Polarized Growth** *Plant Physiol* **176**:352–63
59. Bond LM, Brandstaetter H, Sellers JR, Kendrick-Jones J, Buss F (2011) **Myosin motor proteins are involved in the final stages of the secretory pathways** *Biochem Soc Trans* **39**:1115–9
60. Segal D, Zaritsky A, Schejter ED, Shilo BZ (2018) **Feedback inhibition of actin on Rho mediates content release from large secretory vesicles** *J Cell Biol* **217**:1815–26
61. Rouso T, Schejter ED, Shilo BZ (2016) **Orchestrated content release from Drosophila glue-protein vesicles by a contractile actomyosin network** *Nat Cell Biol* **18**:181–90
62. Olson EN, Nordheim A (2010) **Linking actin dynamics and gene transcription to drive cellular motile functions** *Nat Rev Mol Cell Biol* **11**:353–65
63. Knipe RS, Tager AM, Liao JK (2015) **The Rho kinases: critical mediators of multiple profibrotic processes and rational targets for new therapies for pulmonary fibrosis** *Pharmacol Rev* **67**:103–17
64. Khatiwala CB, Kim PD, Peyton SR, Putnam AJ (2009) **ECM compliance regulates osteogenesis by influencing MAPK signaling downstream of RhoA and ROCK** *J Bone Miner Res* **24**:886–98

- 65. Zhong J, Yang Y, Liao L, Zhang C (2020) **Matrix stiffness-regulated cellular functions under different dimensionalities** *Biomater Sci* **8**:2734–55
- 66. Luo YJ, Su YH (2012) **Opposing nodal and BMP signals regulate left-right asymmetry in the sea urchin larva** *PLoS Biol* **10**
- 67. Siegel S, Castellan N. J. (1988) **Nonparametric statistics for the behavioral sciences 2ed**
- 68. Stalling D, Westerhoff M, Hege H-C (2005) **Amira: A highly interactive system for visual data analysis** *The visualization handbook* **38**:749–67

Article and author information

Eman Hijaze

Department of Marine Biology, Leon H. Charney School of Marine Sciences, University of Haifa, Haifa 31905, Israel

Tsvia Gildor

Department of Marine Biology, Leon H. Charney School of Marine Sciences, University of Haifa, Haifa 31905, Israel

Ronald Seidel

B CUBE Center for Molecular Bioengineering, Technische Universität Dresden, 01309 Dresden, Germany

Majed Layous

Department of Marine Biology, Leon H. Charney School of Marine Sciences, University of Haifa, Haifa 31905, Israel

Mark Winter

Department of Electrical Engineering, Computer Science and Mathematics, Technische Universiteit Delft, Delft 2628CD, Netherlands

ORCID iD: [0000-0003-1180-1957](https://orcid.org/0000-0003-1180-1957)

Luca Bertinetti

B CUBE Center for Molecular Bioengineering, Technische Universität Dresden, 01309 Dresden, Germany

Yael Politi

B CUBE Center for Molecular Bioengineering, Technische Universität Dresden, 01309 Dresden, Germany

Smadar Ben-Tabou de-Leon

Department of Marine Biology, Leon H. Charney School of Marine Sciences, University of Haifa, Haifa 31905, Israel

For correspondence: sben-tab@univ.haifa.ac.il

ORCID iD: [0000-0001-9497-4938](https://orcid.org/0000-0001-9497-4938)

Copyright

© 2023, Hijaze et al.

This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Charles Etensohn

Carnegie Mellon University, Pittsburgh, United States of America

Senior Editor

Kathryn Cheah

University of Hong Kong, Hong Kong, Hong Kong

Reviewer #1 (Public Review):

Using a pharmacological and knock-down approach, the authors could demonstrate that ROCK activity is required for the normal development of the larval skeleton. The presence of ROCK in the pluteus stage depends on the activity of VEGF that is responsible for the formation of the tubular syncytial sheath of the calcifying primary mesenchyme cells in which the skeleton forms. The importance of ROCK in skeleton formation was confirmed in cell culture experiments, demonstrating that ROCK inhibition leads to decreased elongation and abnormal branching of spicules. μ CT analyses underline this finding demonstrating that the inhibition of ROCK mainly affects the elongation of spicules while growth in girth is little affected. F-actin labeling experiments could demonstrate that ROCK inhibition interferes with the organization of the actomyosin network in the early phase of skeleton formation, while f-actin organization in the tips of the elongating spicule is unaffected by the pharmacological inhibition of ROCK. Finally, ROCK inhibition strongly affects the expression of major regulatory and calcification-related genes in the calcifying cells. Based on these findings the authors propose a model for the regulatory interaction between the skeletogenic GRN, ROCK, and the f-actin system relevant for skeletogenesis.

I reviewed this paper previously for submission to another Journal. I emphasize again, that this is an interesting and important work that aims to uncover the interaction between the Rho-associated Kinase, ROCK, the actomyosin network, and its relevance for the formation of the calcitic skeleton of the sea urchin larva. I carefully went through the revised manuscript. In their new version, the authors rearranged the figures to provide a more direct comparison between the in vivo and cell culture experiments which mitigates the criticism of collateral effects by the inhibitors on the whole organism. The authors also performed an additional experiment localizing the F-Actin signal in spicules of PMC cell cultures under ROCK inhibition. This experiment strengthens the concept that ROCK activity is important for tip dominance rather than CaCO_3 deposition rates. The results section was substantially reorganized and only very minor changes were made to the introduction and discussion.

I think that this work has great potential to provide seminal insights into an understudied aspect of the biomineralization process - the role and regulation of the cytoskeleton in calcifying cells. As I mentioned in my previous review there are some gaps in this work that need to be answered to provide a conclusive dataset on the role of ROCK and the actomyosin system in the mineralization process. The manuscript in its current form provides evidence for the interaction of ROCK with the actomyosin system in the sea urchin larva and that perturbation of this system affects skeletogenesis. However, it is missing an explanation

regarding the mechanism by which ROCK affects skeleton formation. No difference in f-actin localization was found at the spicule tips in control and ROCK-inhibited larvae. A slight hint was found in the difference in vesicle size and f-actin organization within calcifying cells, but it remains unresolved if ROCK activity impacts the trafficking of calcification vesicles. The authors provide an interesting discussion on the involvement of f-actin and ROCK on vesicular trafficking, and exocytosis based on existing knowledge from animal and plant models. But for the sea urchin larva, this important link between ROCK, f-actin, and the biomineralization process remains unanswered. In their previous work by Winter et al. 2021, the authors demonstrated excellent technologies to monitor vesicular dynamics in the calcifying cells. This tool would be ideal to investigate the role of ROCK and the actomyosin network on the trafficking dynamics of Ca²⁺-rich vesicles. These experiments (among others suggested in the following review) may help to uncover the critical mechanism to resolve the missing gap in this manuscript.

Major comments

One MASO led to reduced skeleton formation while the other one additionally induced ectopic branching. How was the optimum concentration for the MASOs determined? Did the authors perform a dose-response curve? What is the reason for this difference? Which of the two MASOs can be validated by reduced ROCK protein abundance? Since the ROCK antibody works, I would like to see a control experiment on Rock protein abundance in control and ROCK MO injected larvae which is the gold-standard for validating the knock-down.

L212 "Together, these measurements show that ROCK is not required for the uptake of calcium into cells." But what about trafficking and exocytosis? As mentioned earlier, I think this is a really important point that needs to be confirmed to understand the function of ROCK in controlling calcification. In their previous study (reference 45) the authors demonstrated that they have superior techniques in measuring vesicle dynamics in vivo. Here an acute treatment with the ROCK inhibitor would be sufficient to test if calcein-positive vesicle motion, including the observed reduction in velocity close to the tissue skeleton interface, is affected by the inhibitor.

Is there a colocalization of ROCK and f-actin in the tips of the spicules? This would support the mechano-sensing-hypothesis by ROCK.

L 283. "F-actin is enriched at the tips of the spicules independently of ROCK activity" The results of this paragraph clearly demonstrate that ROCK inhibition has no effect on the localization of f-actin at the tips of the growing spicules. In addition, the new cell culture experiments underline this observation. Still, the central question that remains is, what is the interaction between ROCK, f-actin, and the mineralization process, that leads to the observed deformations? What does the f-actin signal look like in a branched phenotype or in larvae that failed to develop a skeleton (inhibition from Y20)?

Immunohistochemical analyses on f-actin localization and abundance should be additionally performed with ROCK knock-down phenotypes to confirm the pharmacological inhibition.

L 365 "...supporting its role in mineral deposition..." "...Overall, our studies indicate that ROCK activity....is essential for the formation of the spicule cavity.....which could be essential for mineral deposition..." I think the authors need to do a better job in clearly separating between the potential processes impacted by ROCK perturbation. Is it stabilization and mechano-sensing in the spicule tip or the intracellular trafficking and deposition of the ACC? If the dataset does not allow for a definite conclusion, I suggest clearly separating the different possibilities combined with thorough discussion-based findings from other mineralizing systems where the interaction between ROCK and F-actin has been described.

- <https://doi.org/10.7554/eLife.89080.1.sa1>

Reviewer #2 (Public Review):

This manuscript reports on the role of Rho-associated coiled-coil kinase (ROCK) in biomineralization of sea urchin larval skeletons. A number of experiments examine the initiation, growth, and patterning of the skeleton in an effort to determine if, and how, ROCK participates in skeletal formation. The authors conclude that ROCK controls the formation, growth, and morphology (patterning) of the skeleton based on a number of inhibition studies. The main target of the experiments is the actomyosin cytoskeleton which has been the focus of many ROCK studies in vertebrates. Based on similar experimental outcomes when comparing the results here with published data from vertebrates, they suggest that ROCK and the actomyosin network operate in a similar way in biomineralization despite independent evolutionary origins of the sea urchin larval skeletons and the skeletons of vertebrates.

My concerns are the interpretation of the experiments. The main overriding concern is a possible over-interpretation of the role of ROCK. In the literature that ROCK participates in many biological processes with a major contribution to the actin cytoskeleton. And when a function is attributed to ROCK, it is usually based on the determination of a protein that is phosphorylated by this kinase. Here that is not the case. The observation here is in most cases stunted growth of the spicule skeleton and some mis-patterning occurs or there is an absence of skeleton if the inhibitor is added prior to initiation of skeletal growth. They state in the abstract that ROCK impairs the organization of F-actin around the spicules. The evidence for that as a direct role is absent. They use morpholino data and ROCK inhibitor data to draw their conclusion. My main concern is the concentration of the inhibitor used since at the high concentrations used, the inhibitor chosen is known to inhibit other kinases as well as ROCK (PKA and PKC). They indicate that this inhibition is specifically in the skeletogenic cells based on the isolation of skeletogenic cells in culture and spicule production either under control or ROCK inhibition and they observe the same - stunting and branching or absence of skeletons if treated before skeletogenesis commences. Again, however, the high concentrations are known to inhibit the other kinases. They use blebbistatin and latrunculin and show that these known inhibitors of actin cytoskeleton lead to abnormal spiculogenesis. This coincidence is suggestive but is not proof that it is ROCK acts on the actomyosin cytoskeleton given the specificity concerns.

- <https://doi.org/10.7554/eLife.89080.1.sa0>

Author Response:

Thank you very much for selecting our paper for peer review and for the thorough evaluation of our manuscript. We appreciate your assessment and the reviewers' comments that value our work and identify important points that will enable us to improve the paper. We are now working on key experiments to further test the hypothesis that ROCK is essential for the formation, growth, and morphology of the sea urchin larval skeleton. We will address the reviewers' comments in detail in the revised version of the paper that we will submit after completing the experiments, but for now, there are two points we would like to clarify.

We thank the first reviewer for the appreciation of this paper and of our previous work where we studied calcium vesicle dynamics in whole embryos (Winter et al, Plos Com Biol 2021). In Winter et al 2021, we found that the skeleton (spicules) doesn't grow when the embryos are immobilized in either control or treated embryos. As a consequence, we cannot determine the role of ROCK in vesicle trafficking and exocytosis based on experiments conducted in whole embryos. We are developing an alternative assay for vesicle tracking using cell cultures, but that is beyond the scope of this current work.

As for the second reviewer's criticism of the usage of Y-27632 to block ROCK activity: The ROCK inhibitor concentrations we used (30-80 μ M) are similar the those commonly used in

mammalian systems and in *Drosophila* to block ROCK activity, for example: (Becker et al., 2022; Canellas-Socias et al., 2022; Fischer et al., 2009; Kagawa et al., 2022; Segal et al., 2018; Su et al., 2022). The manufactory datasheet indicates that: “Y-27632 dihydrochloride is a selective ROCK inhibitor (Ki values are 0.14-0.22, 0.3, 25, 26 and > 250 μ M for ROCK1 (p160 ROCK), ROCK2, PKA, PKC and MLCK respectively)”. That is, the affinities of Y-27632 for ROCK kinases are at least *100 times higher* than those for PKC, PKA, and MLCK. Furthermore, these Ki values are based on biochemistry assays where the activity of the inhibitor is tested in-vitro with the purified protein. Therefore, these concentrations are not relevant to cell or embryo cultures where the inhibitor has to penetrate the cells and affect ROCK activity in-vivo. Y-27632 activity was studied both in-vitro and in-vivo in Narumiya, Ishizaki and Uehata, *Methods in Enzymology* 2000 (Narumiya et al., 2000). This paper reports similar concentrations to the ones indicated in the manufactory data sheet for the in-vitro experiments, but shows that 10 μ M concentration or higher are effective in cell cultures. As stated above, we will add additional experimental verifications to the revised version, but even at this stage, the concentrations we used and the agreement between our pharmacological and genetic perturbations suggests that the affected protein is indeed ROCK.

We share the reviewers and editors wish to identify the molecular targets of ROCK and the specific cellular processes that ROCK is involved in, and we are actively working on achieving this goal. However, we believe that this paper is an important step towards illuminating the cellular components that participate in biomineral construction and the feedback between the cellular machinery and gene expression.

Best,

Smadar, in the name of all co-authors.

References:

- Becker, K.N., Pettee, K.M., Sugrue, A., Reinard, K.A., Schroeder, J.L., Eisenmann, K.M., 2022. The Cytoskeleton Effectors Rho-Kinase (ROCK) and Mammalian Diaphanous-Related (mDia) Formin Have Dynamic Roles in Tumor Microtubule Formation in Invasive Glioblastoma Cells. *Cells* 11.
- Canellas-Socias, A., Cortina, C., Hernando-Momblona, X., Palomo-Ponce, S., Mulholland, E.J., Turon, G., Mateo, L., Conti, S., Roman, O., Sevillano, M., Slebe, F., Stork, D., Caballe-Mestres, A., Berenguer-Llargo, A., Alvarez-Varela, A., Fenderico, N., Novellademunt, L., Jimenez-Gracia, L., Sipka, T., Bardia, L., Lorden, P., Colombelli, J., Heyn, H., Trepas, X., Tejpar, S., Sancho, E., Tauriello, D.V.F., Leedham, S., Attolini, C.S., Batlle, E., 2022. Metastatic recurrence in colorectal cancer arises from residual EMP1(+) cells. *Nature* 611, 603-613.
- Fischer, R.S., Gardel, M., Ma, X., Adelstein, R.S., Waterman, C.M., 2009. Local cortical tension by myosin II guides 3D endothelial cell branching. *Curr Biol* 19, 260-265.
- Kagawa, H., Javali, A., Khoei, H.H., Sommer, T.M., Sestini, G., Novatchkova, M., Scholte Op Reimer, Y., Castel, G., Bruneau, A., Maenhoudt, N., Lammers, J., Loubersac, S., Freour, T., Vankelecom, H., David, L., Rivron, N., 2022. Human blastoids model blastocyst development and implantation. *Nature* 601, 600-605.
- Narumiya, S., Ishizaki, T., Uehata, M., 2000. Use and properties of ROCK-specific inhibitor Y-27632. *Methods Enzymol* 325, 273-284.
- Segal, D., Zaritsky, A., Schejter, E.D., Shilo, B.Z., 2018. Feedback inhibition of actin on Rho mediates content release from large secretory vesicles. *J Cell Biol* 217, 1815-1826.
- Su, Y., Huang, H., Luo, T., Zheng, Y., Fan, J., Ren, H., Tang, M., Niu, Z., Wang, C., Wang, Y., Zhang, Z., Liang, J., Ruan, B., Gao, L., Chen, Z., Melino, G., Wang, X., Sun, Q., 2022. Cell-in-cell structure mediates in-cell killing suppressed by CD44. *Cell Discov* 8, 35.